

8(a)	microvilli ; A brush border	1
8(b)	no movement of, chemicals / substances / ions, between the, chemoreceptor / cells / (cells) A ; Ignore food A named dissolved substance	1

Question	Answer	Marks
8(c)	<i>any seven from:</i> 1. Na⁺ ions enter, (chemo)receptor cell / A, through, microvilli / Y / channel (proteins) ; 2 cell surface membrane depolarised ; 3 receptor potential ; 4 <i>ref. to</i> threshold ; 5 (voltage-gated) Ca²⁺ / calcium, channels open ; 6 Ca²⁺ ions enter, cytoplasm / cell / A ; A presynaptic knob 7 vesicles (of neurotransmitter), move towards / fuse with, cell surface membrane / presynaptic membrane ; 8 exocytosis (described) / secretion, of (named) neurotransmitter ; 9 neurotransmitter binds to receptor on, postsynaptic / (sensory) neurone / B, <u>membrane</u> ; 10 Na⁺ / sodium, channels open or Na⁺ enter (sensory) neurone / B ; 11 postsynaptic membrane / (sensory) neurone membrane, depolarised ; 12 <i>ref. to</i> threshold ; <i>allow either mp4 or mp12</i>	7